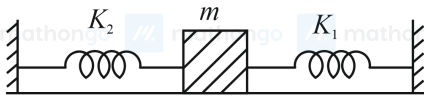


1. A mass m is attached to two springs as shown in figure. The spring constants of two springs are K_1 and K_2 . For the frictionless surface, the time period of oscillation of mass m is



[2023 (06 Apr Shift 1)]

- (1) $2\pi\sqrt{\frac{m}{K_1+K_2}}$
 - (2) $\frac{1}{2\pi}\sqrt{\frac{K_1-K_2}{m}}$
 - (3) $2\pi\sqrt{\frac{m}{K_1-K_2}}$
 - (4) $\frac{1}{2\pi}\sqrt{\frac{K_1+K_2}{m}}$
2. A small block of mass 100 g is tied to a spring of spring constant 7.5 N m^{-1} and length 20 cm. The other end of spring is fixed at a particular point A . If the block moves in a circular path on a smooth horizontal surface with constant angular velocity 5 rad s^{-1} about point A , then tension in the spring is

[2023 (06 Apr Shift 1)]

- (1) 0.75 N
 - (2) 0.25 N
 - (3) 0.50 N
 - (4) 1.5 N
3. At any instant the velocity of a particle of mass 500 g is $(2t \hat{i} + 3t^2 \hat{j}) \text{ m s}^{-1}$. If the force acting on the particle at $t = 1 \text{ s}$ is $(\hat{i} + x \hat{j}) \text{ N}$. Then the value of x will be:

[2023 (08 Apr Shift 1)]

- (1) 3
 - (2) 4
 - (3) 2
 - (4) 6
4. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**
- Assertion A** : An electric fan continues to rotate for some time after the current is switched off.
- Reason R** : Fan continues to rotate due to inertia of motion.
- In the light of above statements, choose the most appropriate answer from the options given below.

[2023 (10 Apr Shift 2)]

- (1) **A** is correct but **R** is not correct
 - (2) **A** is not correct but **R** is correct
 - (3) Both **A** and **R** are correct and **R** is the correct explanation of **A**
 - (4) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**
5. A coin placed on a rotating table just slips when it is placed at a distance of 1 cm from the centre. If the angular velocity of the table is halved, it will just slip when placed at a distance of _____ from the centre:

[2023 (11 Apr Shift 1)]

- (1) 8 cm
 - (2) 4 cm
 - (3) 1 cm
 - (4) 2 cm
6. A body of mass 500 g moves along x -axis such that its velocity varies with displacement x according to the relation $v = 10\sqrt{x} \text{ m s}^{-1}$ the force acting on the body is:

[2023 (11 Apr Shift 2)]

- (1) 125 N
- (2) 25 N
- (3) 166 N
- (4) 5 N

7. The position vector of a particle related to time t is given by $\vec{r} = (10t\hat{i} + 15t^2\hat{j} + 7t\hat{k})$ m. The direction of net force experienced by the particle is :

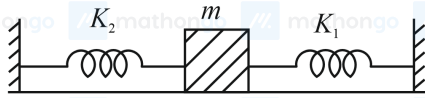
[2023 (15 Apr Shift 1)]

- (1) Positive x -axis
- (2) In $x - y$ plane
- (3) Positive y -axis
- (4) Positive z -axis

ANSWER KEYS

1. (1) 2. (1) 3. (1) 4. (3) 5. (2) 6. (2) 7. (3)

1. (1)



For the given combination, if the block is displaced towards right, elongation in one spring will be equal to the compression in other and the direction of force due to both spring will be the same and restoring in nature. Hence, both springs are effectively in parallel combination.

Therefore,

$$\Rightarrow K_{eff} = K_1 + K_2 \text{ and}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{m}{K_{eff}}} = 2\pi\sqrt{\frac{m}{K_1 + K_2}}$$

2. (1)

Centripetal force will be provided by the spring force. Let the elongation in the spring be x , then we can write

$$\Rightarrow kx = m\omega^2(r + x)$$

$$\Rightarrow 7.5x = 2.5(0.2 + x)$$

$$\Rightarrow x = \frac{0.5}{5} = 0.1$$

Therefore, required tension will be $T = kx = 0.75 \text{ N}$.

3. (1)

The equation for velocity is

$$\vec{v} = (2t \hat{i} + 3t^2 \hat{j})$$

Thus, the acceleration is

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d(2t \hat{i} + 3t^2 \hat{j})}{dt} = (2 \hat{i} + 6t \hat{j})$$

$$\text{At } t = 1 \text{ s, } \vec{a} = 2 \hat{i} + 6 \hat{j}$$

$$\text{The force is written as } \vec{F} = m\vec{a} = \frac{2\hat{i} + 6\hat{j}}{2} = \hat{i} + 3\hat{j}$$

Comparing it with the given value of force, $x = 3$.

4. (3)

When a fan is rotating, it acquires some kinetic energy. When the switch is off, because of its motion, it continues rotating for some time until it stops because of the air friction. So it is due to inertia of motion that fan continues to rotate.

Hence, both **A** and **R** are correct and **R** is the correct explanation of **A**.

5. (2)

The coin will slip when

$$\mu mg = m\omega^2 r$$

$$\Rightarrow \mu g = \omega^2 \times 1 \text{ cm} \dots (i)$$

$$\text{Now, } \omega' = \frac{\omega}{2}$$

$$\mu g = \left(\frac{\omega}{2}\right)^2 R \dots (ii)$$

Hence we can write from (i) and (ii)

$$\omega^2 \times 1 \text{ cm} = \frac{\omega'^2}{4} R$$

$$\Rightarrow R = 4 \text{ cm}$$

6. (2)

It is given that

$$v = 10\sqrt{x} \text{ m s}^{-1}$$

Force is defined by $F = m \frac{dv}{dt} \dots (i)$

Multiplying and dividing equation (i) by dx

$$F = m \frac{dv}{dt} \frac{dx}{dx}$$

$$\Rightarrow F = m \frac{dx}{dt} \frac{dv}{dx} = mv \frac{dv}{dx} \dots (ii)$$

Finding the values of the terms in equation (ii)

$$\frac{dv}{dx} = \frac{10}{2\sqrt{x}} = \frac{5}{\sqrt{x}}$$

$$v \frac{dv}{dx} = a = 10\sqrt{x} \times \frac{5}{\sqrt{x}} = 50 \text{ m s}^{-2}$$

Hence, the force is

$$F = \frac{1}{2} \times 50 = 25 \text{ N.}$$

7. (3)

Given, the position vector of the particle is

$$\vec{r} = 10t\hat{i} + 15t^2\hat{j} + 7\hat{k}$$

The velocity of the particle can be calculated as follows:

$$\begin{aligned} \vec{v} &= \frac{d\vec{r}}{dt} \\ &= \frac{d}{dt}(10t\hat{i} + 15t^2\hat{j} + 7\hat{k}) \\ &= 10\hat{i} + 30t\hat{j} \end{aligned}$$

And, the acceleration of the particle is given by

$$\begin{aligned} \vec{a} &= \frac{d\vec{v}}{dt} \\ &= \frac{d}{dt}(10\hat{i} + 30t\hat{j}) \\ &= 30\hat{j} \end{aligned}$$

Hence, the force on the particle can be written as

$$\begin{aligned} \vec{F} &= m\vec{a} \\ &= 30m\hat{j} \end{aligned}$$